NOVATECH

HR Attrition Analytics Dashboard

PROJECT METHODOLOGY

A documented end-to-end analytical workflow for data preparation, validation, KPI development, dashboard construction, and business interpretation.

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Microsoft Excel Portfolio Project

Document Control

Methodology governance and navigation

Field	Detail
Document title	Project Methodology - HR Attrition Analytics Dashboard
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Company	NovaTech (fictional portfolio organization)
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Author / analyst	Amirreza Akbari
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Related files	Business Requirements, Data Dictionary, Executive Report, Validation Checklist

Revision History

Version	Date	Owner	Change
0.1	July 2026	Amirreza Akbari	Initial methodology structure
0.9	July 2026	Amirreza Akbari	Technical content and controls reviewed
1.0	July 2026	Amirreza Akbari	Final portfolio release

Contents

1. Purpose and Analytical Principles

How the analysis was structured and governed

This methodology documents the repeatable process used to transform NovaTech employee records into an executive HR attrition dashboard. It connects the business requirements to the technical implementation and provides enough detail for another analyst to review, validate, and reproduce the work.

Methodological objective: produce accurate, traceable, decision-ready attrition analytics without altering the original source data.

Core Analytical Principles

Principle	Application in this project
Business-first analysis	Every calculation and chart addresses a defined HR question or retention decision.
Source preservation	Raw_Data remains unchanged and acts as the project audit trail.
Controlled transformation	Cleaning and helper fields are created only in Cleaned_Data.
Metric consistency	Attrition rates use a documented numerator and denominator across all views.
Traceability	Dashboard elements link back to PivotTables and defined fields.
Association, not causation	Findings describe observed patterns; they do not claim causal proof.
Audience-centered design	The dashboard is optimized for rapid interpretation by HR managers and executives.

Analytical Approach

- Descriptive analytics to quantify workforce composition and attrition.
- Diagnostic segmentation to compare attrition across departments, roles, demographics, work conditions, compensation, satisfaction, and tenure.
- Interactive exploration through PivotTables, PivotCharts, and slicers.
- Business translation through written insights and prioritized recommendations.

2. End-to-End Workflow

From business question to management action

Stage	Activity	Control / Output
1	Business framing	Define the problem, users, KPIs, scope, and acceptance criteria.
2	Data intake	Load the source employee dataset and preserve the original records.
3	Data profiling	Review row counts, fields, formats, uniqueness, blanks, and value ranges.
4	Cleaning	Remove invalid or duplicate records and standardize analytical fields.
5	Feature engineering	Create analysis-ready helper fields and business groupings.
6	Validation	Reconcile record counts, formulas, categories, and KPI outputs.
7	Aggregation	Build PivotTables for each business question and segment.
8	Visualization	Convert validated summaries into clear executive charts and KPI cards.
9	Interpretation	Document evidence-based findings, limitations, and HR implications.
10	Handover	Package the workbook, documentation, and repository for reuse.

Stage Gates

Gate	Completion condition
G1 - Data readiness	Raw records preserved; analytical table structured; fields documented.
G2 - Quality approval	Duplicates, blanks, data types, and allowed values checked.
G3 - Metric approval	Headcount, attrition count, active employees, and attrition rate reconciled.
G4 - Dashboard approval	Charts respond correctly to slicers and show percentage-based comparisons.
G5 - Reporting approval	Insights are supported by data and recommendations are linked to findings.

3. Data Intake and Workbook Architecture

Separation of source, transformation, analysis, and presentation

The final workbook uses seven worksheets. This layered structure reduces accidental changes, improves traceability, and makes the project easier to review.

Worksheet	Role in methodology	Key control
Raw_Data	Original imported employee records.	No transformation or deletion after intake.
Cleaned_Data	Validated records plus derived analytical fields.	One row per employee; consistent formats and categories.
Data_Dictionary	Definitions for fields used in the project.	Aligns business meaning, type, and permitted values.
Pivot_Tables	Validated aggregations supporting KPIs and charts.	Rates use segment-specific denominators.
Dashboard	Executive presentation layer.	Connected to approved PivotTables and slicers.
Insights	Written interpretation of observed patterns.	Each statement must be supported by a visible result.
Recommendations	Proposed HR actions linked to findings.	Actions are prioritized and framed as decisions, not guarantees.

Dataset Profile

Measure	Observed structure
Raw data	1,488 employee rows across 35 source fields
Cleaned data	Approximately 1,474 analysis-ready rows across 37 fields
Derived fields	AgeGroup, AttritionFlag, SatisfactionGroup, IncomeGroup, TenureGroup
Target outcome	Attrition (Yes / No)
Primary identifier	EmployeeNumber
Analysis grain	One record per employee

The workbook separates raw and cleaned data so that every analytical result can be traced back to the source while preserving an auditable baseline.

4. Data Cleaning and Preparation

Controls applied before analysis

Data preparation focused on creating a reliable analytical table while retaining the original source. Cleaning decisions were designed for a descriptive dashboard, not for predictive modeling.

Control	Method	Acceptance rule
Duplicate employee check	Evaluate EmployeeNumber uniqueness and duplicate rows.	One analytical row per employee.
Blank-value review	Filter and count blanks across all source fields.	No unexplained blanks in fields used for KPIs or segmentation.
Data-type review	Confirm numeric, categorical, and identifier fields.	Ages, income, rates, scores, and tenure stored as numeric values.
Category standardization	Normalize text labels and remove leading/trailing spaces.	Equivalent categories use one spelling and capitalization.
Range validation	Check age, satisfaction scales, income, percentages, and years.	Values fall within documented business ranges.
Constant-field handling	Exclude non-informative constant fields from analytical views.	Raw source remains intact; analysis focuses on variable fields.
Record disposition	Remove only records failing documented validity or duplication rules.	Raw-to-clean difference is explainable and reviewable.

Preparation Sequence

1. Copy valid source records into the Cleaned_Data analytical table.
2. Standardize column naming and arrange fields by business topic.
3. Validate EmployeeNumber and preserve it as the row-level identifier.
4. Review numerical values and categorical labels.
5. Add helper fields using documented formulas and thresholds.
6. Convert the final range into a structured Excel Table for consistent formulas and PivotTable sourcing.

Fields not used as analytical dimensions

Source fields with no variation or no decision value may be retained in Raw_Data but excluded from dashboard analysis. The identifier is used for counting and validation, not as a business category.

5. Data Validation and Quality Assurance

Evidence that the workbook is analysis-ready

Validation was performed at three levels: record-level integrity, field-level validity, and output-level reconciliation.

Validation level	Checks	Evidence / result
Record integrity	Row counts, duplicate identifiers, duplicate records.	Cleaned row count reconciled to removed invalid or duplicate records.
Field completeness	Blank checks in analysis-critical fields.	No unexplained blanks in target or KPI fields.
Categorical validity	Attrition, department, role, gender, overtime, and status labels.	Values match documented categories.
Numerical validity	Age, income, satisfaction, work-life balance, tenure, and percentages.	Values remain within realistic and documented ranges.
Formula validity	Helper-column formulas reviewed for boundary conditions.	Every employee maps to one valid group and flag.
Aggregation validity	Pivot counts and sums compared with direct formulas.	Headcount and attrition totals reconcile.
Dashboard validity	Slicer behavior, chart links, labels, and percentage formats.	Filters update all intended visuals consistently.

Key Reconciliation Tests

- Total Employees = count of unique valid EmployeeNumber values.
- Attrition Count = sum of AttritionFlag and count of Attrition = Yes.
- Active Employees = Total Employees - Attrition Count.
- Overall Attrition Rate = Attrition Count / Total Employees.
- For every segment: employee counts across categories sum to the overall headcount.
- Segment attrition rates use employees in the same segment as the denominator.

Counts alone are not used to compare differently sized groups. Department, role, demographic, and work-condition comparisons are evaluated with attrition rates.

6. Feature Engineering

Converting source fields into decision-friendly segments

Derived field	Logic	Analytical purpose
AttritionFlag	Yes = 1; No = 0	Enables sums, rates, and PivotTable calculations.
AgeGroup	Under 25; 25-34; 35-44; 45-54; 55+	Compares retention patterns across career stages.
SatisfactionGroup	1 = Low; 2 = Medium; 3 = High; 4 = Very High	Translates coded ratings into readable categories.
IncomeGroup	Income thresholds defined in the workbook	Supports compensation-related segmentation.
TenureGroup	0-1; 2-4; 5-9; 10+ years	Highlights onboarding and early-tenure retention risks.

Design Rules for Derived Fields

- Categories are mutually exclusive: one employee belongs to exactly one group.
- Categories are collectively exhaustive: every valid employee receives a group.
- Thresholds are documented in the Data Dictionary and remain consistent across PivotTables.
- Labels are ordered logically so charts read from lower to higher values or shorter to longer tenure.
- Derived fields simplify interpretation without replacing the original numeric source fields.

Formula Governance

Helper columns use structured references within the Cleaned_Data Excel Table. Formulas are filled consistently down the entire table, reviewed at boundary values, and checked for blank or unexpected outputs. The source fields remain available for deeper analysis and validation.

7. KPI and PivotTable Development

Metric definitions and analytical summaries

The project uses a small set of executive KPIs and supporting PivotTables. Each KPI has one definition and one source of truth.

KPI	Definition	Validation
Total Employees	Count of valid employee records / unique EmployeeNumber.	Reconcile table rows and PivotTable grand total.
Attrition Count	Employees where Attrition = Yes; equivalent to Sum of AttritionFlag.	COUNTIF equals PivotTable sum.
Active Employees	Employees where Attrition = No or Total Employees minus Attrition Count.	Independent formula cross-check.
Attrition Rate	Attrition Count divided by Total Employees.	Displayed as percentage with consistent rounding.
Average Age	Mean Age across the filtered population.	Compare formula and PivotTable average.
Average Monthly Income	Mean MonthlyIncome across the filtered population.	Compare formula and PivotTable average.

PivotTable Analysis Set

View	Rows / grouping	Primary measure
Department	Department	Employees, attrition count, attrition rate
Job role	JobRole	Employees, attrition count, attrition rate
Overtime	OverTime	Employees and attrition rate
Age	AgeGroup	Employees and attrition rate
Income	IncomeGroup	Employees and attrition rate
Satisfaction	SatisfactionGroup / JobSatisfaction	Employees and attrition rate
Tenure	TenureGroup	Employees and attrition rate
Demographics	Gender / MaritalStatus	Employees and attrition rate

8. Dashboard Design and Interactivity

How validated analysis becomes an executive decision tool

The Dashboard sheet translates PivotTable outputs into a compact executive view. Visual hierarchy is used so the user can understand the situation, identify the strongest patterns, and explore segments without reviewing the underlying tables.

Dashboard zone	Content	Design intention
Header	Project title, subtitle, reporting context.	Establish purpose immediately.
KPI row	Headcount, attrition count, active employees, attrition rate, averages.	Communicate scale and overall status in seconds.
Primary analysis	Department, job role, overtime, and age group.	Surface the most decision-relevant risk patterns.
Secondary analysis	Income, satisfaction, tenure, work-life balance, and demographics.	Enable deeper diagnostic exploration.
Slicers	Department, role, gender, age group, income group, overtime, marital status.	Allow non-technical users to segment the workforce.
Narrative support	Insights and recommendations sheets.	Separate evidence from proposed action.

Visualization Standards

- Use two-dimensional bar, column, and line charts; avoid 3D effects.
- Use consistent colors, with the attrition emphasis color reserved for risk or departure metrics.
- Display percentages where group sizes differ.
- Sort categories to support comparison, not alphabetically when ranking is more useful.
- Use direct labels where they improve clarity and avoid unnecessary legends.
- Keep chart titles business-focused, such as “Attrition Rate by Department.”
- Ensure all intended PivotCharts respond to the relevant slicers.

Usability Testing

The dashboard is tested by clearing filters, applying one slicer at a time, applying combinations of slicers, and confirming that KPI values, charts, titles, and labels remain coherent. Extreme filters with small populations are reviewed to avoid misleading interpretation.

9. Insight Generation and Recommendations

Turning patterns into defensible business guidance

Analytical outputs were translated into written findings using a consistent evidence structure. This prevents unsupported conclusions and keeps the reporting focused on business decisions.

Element	Required content	Example form
Observation	What the metric or chart shows.	"The overtime group has a higher observed attrition rate."
Evidence	The relevant rate, comparison, or ranking.	Include the filtered percentage or difference.
Business implication	Why the pattern matters to HR.	Potential workload, scheduling, or burnout exposure.
Limitation	What the data cannot prove.	Association does not establish overtime as the cause.
Recommended action	A practical response linked to the evidence.	Review workload allocation and monitor overtime-intensive teams.

Recommendation Prioritization

Priority lens	Question used
Impact	Could the action materially reduce attrition or protect a critical workforce segment?
Evidence strength	Is the recommendation supported by a clear and repeatable pattern?
Feasibility	Can HR or line management implement the action with available resources?
Specificity	Is the target group, owner, and intended result clear?
Measurability	Can the organization track the result through future dashboard refreshes?

Recommendations are framed as testable management actions. The dashboard supports prioritization and monitoring; it does not replace employee research, manager interviews, or causal analysis.

10. Reproducibility, Limitations, and Handover

How the project can be reviewed, reused, and extended

Reproducibility and Handover

7. Retain Raw_Data as the unchanged source layer.
8. Replace or append source records only through a controlled update process.
9. Apply the same cleaning and validation rules to new records.
10. Confirm helper formulas extend to all new rows.
11. Refresh PivotTables and verify all slicer connections.
12. Reconcile KPI totals before publishing a refreshed dashboard.
13. Update Insights and Recommendations only after reviewing the refreshed evidence.
14. Maintain version history for the workbook and supporting PDFs.

Known Limitations

- The dataset is a portfolio dataset and may not represent the full complexity of a live HR information system.
- The analysis is cross-sectional and descriptive; it does not establish causation or time-to-event behavior.
- No predictive attrition model, confidence interval, or statistical significance test is included.
- Income and satisfaction findings may be influenced by role, level, department, or other confounding factors.
- The workbook is not connected to a live HR system and requires controlled manual refresh.
- Privacy, access controls, and employee-level confidentiality would require stronger governance in a production environment.

Future Enhancements

- Add monthly snapshots and trend analysis.
- Introduce cohort retention and early-tenure survival views.
- Build a Power BI version with role-based access and automated refresh.
- Add statistical testing and multivariable analysis.
- Track retention initiatives and compare pre/post intervention results.
- Create a predictive model only after ethical, legal, and governance review.

Final Deliverables

Deliverable	Purpose
Excel workbook	Source, cleaning, calculations, PivotTables, dashboard, insights, and recommendations.
Business Requirements PDF	Defines the business need, scope, requirements, and acceptance criteria.
Data Dictionary PDF	Defines fields, values, derived logic, and data-quality expectations.
Project Methodology PDF	Documents the workflow and analytical controls.
Executive Report PDF	Summarizes results and recommended actions for leadership.
Portfolio Case Study PDF	Communicates the project story and demonstrated skills to recruiters.
Validation Checklist PDF	Provides final QA evidence before publication.

Appendix A - Methodology Traceability Matrix

Connection between business questions, analysis, and outputs

Business question	Primary fields	Method	Output
What is the overall attrition rate?	Attrition, AttritionFlag, EmployeeNumber	Count and rate reconciliation	KPI cards
Which departments are highest risk?	Department, AttritionFlag	Department-level rates	Bar chart and insight
Which roles are most affected?	JobRole, AttritionFlag	Role-level rates and ranking	Horizontal bar chart
Does overtime relate to attrition?	OverTime, AttritionFlag	Two-group rate comparison	Column chart
Which age groups leave more?	Age, AgeGroup, AttritionFlag	Grouped rate comparison	Age-group chart
Does income relate to attrition?	MonthlyIncome, IncomeGroup, AttritionFlag	Band-based comparison	Income-group chart
How does satisfaction relate to attrition?	JobSatisfaction, SatisfactionGroup, AttritionFlag	Ordinal group comparison	Satisfaction chart
When is tenure risk highest?	YearsAtCompany, TenureGroup, AttritionFlag	Tenure-band comparison	Tenure chart
What should HR prioritize?	All validated analytical outputs	Evidence-impact-feasibility review	Recommendations sheet and executive report

Appendix B - Glossary

Term	Definition
Attrition	An employee record marked as having left the organization.
Attrition rate	Employees who left divided by employees in the relevant population.
Headcount	Number of valid employee records in the selected population.
Derived field	A new analytical variable calculated from one or more source fields.
PivotTable	Excel summary structure used to aggregate and segment the dataset.
Slicer	Interactive Excel filter connected to one or more PivotTables.
KPI	A high-level metric used to monitor workforce status or performance.
Reconciliation	Independent comparison confirming that totals and calculations agree.